

Electricity supply and demand scenarios for the Southern African power pool



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ABSTRACT

The study presents long-term electricity supply and demand scenarios for the twelve countries in the Southern African Power Pool, based on detailed bottom-up demand analysis for all countries and a set of internally consistent development scenarios. Total regional electricity demand and supply increase by eight to fourteen times from 2010 to 2070, with major shifts in both the sectoral composition of demand and the geography of demand, with South Africa becoming a much smaller share. On the supply side, the fuel mix shifts from coal and toward hydro in the medium term, but towards other renewables, such as solar, in the longer term, particularly in the scenarios with the fastest decline in capital costs for renewables. This leads to declining unit carbon dioxide emissions in the more aggressive scenarios, even though total power sector emissions still increase. The unit cost of generation for the entire region is stable across all scenarios. The potential transformation of the supply sector would require a fundamental shift in resource use, grid management and infrastructure development in the region, as well as greater regional integration. This also implies significant institutional capacity development in the SAPP Coordination Centre or similar structures for cooperative management of resources.

1. Introduction

The southern African region has experienced sustained economic growth and increasing prosperity over the past decade, driven largely by increasing demand for natural resource-based commodities and facilitated by increased peace and stability. As the Southern African Development Community (SADC) region industrialises on its path to improved human development, the demand for power is increasing dramatically. As a result, the electricity sector is a key component of the infrastructure that drives both regional integration and economic growth, with energy security being increasingly important to continued

development across southern Africa (AU, 2012; Eberhard et al., 2011). At the same time, the chronic power shortages in the region in recent years has hampered short-term economic development. The Southern African Power Pool (SAPP), established in 1995, provides a forum for regional solutions to electricity generation and supply through coordinated planning and operation of the regional power system, which consists of generators and international inter-connectors.

Since the early days of SAPP, numerous studies have examined the outlook for power sector expansion in the region, as well as the potential benefits from increased trade and cooperation on regional projects (Alfstad, 2005; Bowen et al., 1999; Economic Consulting

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Associates, 2009; Nexant, 2007; Rowlands, 1998). More recently, two studies have looked in more detail at the role of renewable energy in the development of the SAPP system – the SADC Renewable Energy Strategy and Action Plan (RESAP) (CEEZ, 2012) and a study by the Energy Research Centre and the International Renewable Energy Agency (IRENA) (Miketa and Merven, 2013). In addition, the SAPP Coordination Centre compiles the demand and supply forecasts from the national utility members and publishes this ten-to-fifteen-year outlook each year, although without any further analysis (e.g. SAPP, 2014, 2013). While these studies often provide detailed supply optimisation analysis, none of them include detailed bottom-up demand analysis. In fact, many of the studies either rely on utility estimates (which are rarely based on bottom-up analysis) or simply use a constant annual growth rate over the study period. In addition, the time frame for most studies was limited to 20 years, or even 10 years for the SAPP reports (CEEZ, 2012; Economic Consulting Associates, 2009; Miketa and Merven, 2013; Nexant, 2007; SAPP, 2014). Even the recent IRENA study, which extended the timeframe to 2050, simply used an extrapolation of earlier national growth rates for this longer period. One additional study that did include bottom-up demand analysis out to 2030 (Merven et al., 2010), did not include any supply analysis. A 20-year timeframe for analysis has two important limitations: first, the declining costs of renewable power alternatives may take several decades to tip the balance away from fossil fuel dependence in supply planning, as well as to address the limited opportunities for storing non-dispatchable power. Second, many of the SAPP national power systems are dominated by hydropower, which could be vulnerable to long-term changes in climate – and therefore water availability – but this impact may only be visible over 30–50 years (Spalding-Fecher et al., 2014; Stanzel and Kling, 2014). A final important issue with earlier studies is that the underlying drivers of electricity demand, such as population growth, economic growth and the shifts in the structure of the economy, are not always internally consistent or do not have a coherent development framework. This makes it difficult to compare the results, because the underlying visions of the future may be quite different from study to study, and this is not made explicit in those estimates.

The objective of this study is to analyse and provide projections of electricity supply and demand for SAPP over a long time period (2010–2070), based on a set of internally consistent development scenarios, and using bottom-up demand analysis. In addition, the analysis combines a simulation of the stated expansion plans of the regional electricity utilities (e.g. out to 2025) with an optimisation analysis of what power options should be used to meet demand over the longer term, taking into consideration changing capital and fuel costs. Section 2 introduces the energy modelling framework and overall methodology. Section 3 then presents data and assumptions used for modelling demand, supply, trade, generation costs, and greenhouse gas emissions, as well as the calibration of the modelling framework. The demand and supply results are then presented in Section 4, along with discussion, followed by conclusions in Section 5.

2. Methodology

As Bazilian et al. (2012) and Koppelaar et al. (2016) explain, there are numerous long-term energy forecasting and simulation modelling tools that each have their own strengths and weakness. The tool selected for this analysis is the Long-range Energy Alternatives Planning (LEAP) modelling system, developed by Stockholm Environment Institute (SEI) (Heaps, 2012), and increasingly used as part of integrated water-energy-climate modelling analyses (see, e.g., Howells et al., 2013; Sattler et al., 2012; Yates and Miller, 2013).

The overall structure of LEAP is presented in Fig. 1 showing the main flows of information through LEAP for this analysis. LEAP is not a model of a particular energy system; rather it a flexible software framework within which models of different energy systems can be

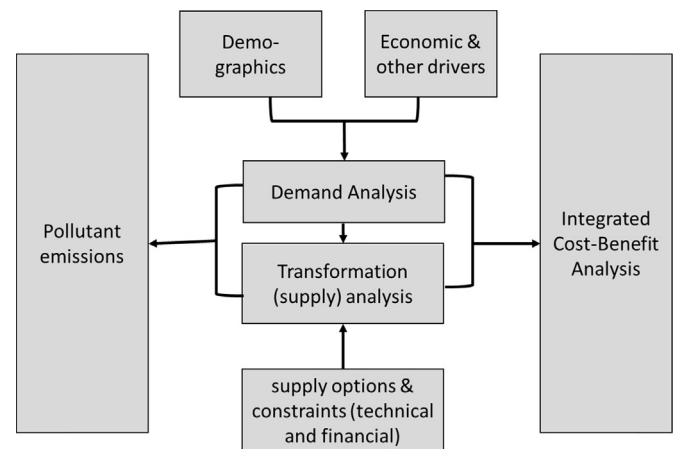


Fig. 1. Structure of LEAP model as applied in this study.

constructed. Its most important features for this study are its support for multi-regional (in this case multi-country) analyses, alternative scenario projections, and the ability to combine bottom-up energy service-based demand forecasts with least-cost optimisation modelling of electric generation.

LEAP models are demand-driven and typically combine bottom-up energy service-based energy demand forecasts with simulation or optimisation-based models of energy production and conversion (which in LEAP is referred to as “Transformation”). LEAP’s demand models are based around a straightforward accounting approach that calculates energy consumption as the product of some type of activity level and an annual average energy intensity specified as units of energy consumption per unit of activity. Activity levels are typically broken down into their various components within a hierarchical tree structure displayed within LEAP and used to organize the main sources of data. For example, in the household sector energy intensities may be specified per household by fuel for each major end-use (cooking, lighting, appliances, etc.), while the total number of households in each country may be broken down into urban versus rural households and then into electrified and unelectrified households. The user is free to specify how each of these values may evolve in the future based, for example, on expected rates of population growth, urbanisation, electrification and technology penetration. In the industry, services and agriculture sectors, energy consumption can be disaggregated by major subsectors, and energy intensities may be specified per unit of value added in each subsector. LEAP models are typically used for integrated energy planning that considers all fuels and the potential for substitution among fuels and technologies. However, for this study, the demand modelling is limited to consider only demands for electricity. The major macroeconomic and demographic assumptions used in the study are described in details in Sections 3.1 and 3.2.

In terms of the Transformation analysis, the model developed for this study combines a relatively simple set of accounting projections for transmission, distribution and own use energy losses, with a multi-regional least-cost optimisation model for electric generation. LEAP’s optimisation calculations are based on the Open Source Energy Modelling System (OSeMOSYS) (Howells et al., 2011) and the GNU Linear Programming Kit (GLPK), a software toolkit intended for solving large scale linear programming problems by means of the revised simplex method and the CPLEX Solver. This system can be used to calculate least-cost pathways for capacity expansion and plant dispatch in any particular scenario. The assumptions behind the supply analysis are presented in Section 3.3.

LEAP can also be used to calculate the emissions of greenhouse gases and other local air pollutants in any scenario through the specification of emissions factors, typically entered as emissions per unit of energy combusted. LEAP’s optimisation calculations can use an

Table 1
Summary of development scenarios.
Source: Authors' analysis

	BAU	SADC integration	Grand Deal
GDP per capita growth	Low	Med	High
Population growth	Highest	Lower	Lowest
Hydropower investment	Slower	Faster	Much faster
Other Clean energy investment	Limited	Some	Rapid
Investment in energy intensive industry	Current trends	Faster	Faster, but with better technology
Regional trade	Limited	Large	Large
Grand Inga	Much later	Later	Soon
Technology learning for renewable energy	Moderate	Moderate	Fast
Improvement in industrial energy intensity	None	Moderate	Fast

objective function to minimize total economic cost for the entire electricity system. The capability of coupling the electricity modelling with a water modelling system (i.e. SEI's WEAP model) means that the climate change impacts on the power system can be analysed simultaneously in future research.

The electricity system model presented here covers the following countries: Angola, Botswana, Democratic Republic of Congo (DRC), Lesotho, Malawi, Mozambique, Namibia, South Africa, Swaziland, Tanzania, Zambia, and Zimbabwe. Even though Angola, Malawi and Tanzania are not currently connected to the SAPP regional grid, there are inter-connector projects currently underway. The system boundary for costs in this study is electricity generation only (i.e. not demand-side costs and no transmission and distribution costs) and covers only the electricity sector (supply and demand) and the fuel inputs for power generation. It does not address primary energy sources used for non-electric end uses (e.g. biomass) or the use of other fuels directly in electricity consuming sectors (e.g. fuel use in boilers). Power plants are grouped into existing plants, specific plants (i.e. new planned projects where the project site is specified), and generic plants (i.e. ones that could be built anywhere in the country). The flexibility and scalability of the modelling system easily accommodates the 12 continental countries of SADC and more than 200 power plant options.

The structure for demand modelling for SAPP countries has been structured around the primary economic sectors used in the World Development Indicators database (agriculture, manufacturing, other industry, and services),³ plus the residential sector (disaggregated into rural and urban households). This structure also matches the main International Energy Agency (IEA) categories for electricity demand (IEA, 2014a), with the IEA's "mining and quarrying" being comparable to the WDI's "other industry", which is called "extractive" in this study. A key assumption in the application of LEAP is that demand is not constrained by a lack of infrastructure and supply shortages. In the short term however, widespread shortages will persist in reality and these will negatively affect economic development. The purpose of this analysis, therefore, is to illustrate what expansion of generating and distribution capacity would be required to meet likely future demands for development, if they were not constrained by investment or other supply constraints. Because transportation sector consumption of electricity is almost non-existent in most SAPP countries, this was not included in the modelling except in South Africa, where electric rail for passengers and freight is included (ERC, 2013).

³ The WDI reports "industry" as a percentage of GDP, and "manufacturing" as a subset of industry. The difference between total industry, and manufacturing, which is "other manufacturing", is what is called "extractive" in this study because it includes mining, quarrying and oil extraction.

3. Data

3.1. Development scenarios

A critical methodological issue for long-term scenario planning is creating scenarios that have internally consistent storylines linked by descriptions of key economic and social systems (Kahane, 2012, 2000; Shell, 2001; Van der Heijden, 1996). We use three scenarios in this analysis, called "Business as Usual", "SADC Integration" and "Grand Deal". The IPCC has developed a new approach to scenario development, called "shared socio-economic pathways" (SSPs), which includes qualitative and quantitative aspects of future development, including the policy responses to climate change (Arnell et al., 2011; Nakicenovic et al., 2014; O'Neill et al., 2014; Vuuren and Carter, 2014). "Business as Usual" is analogous to the IPCC SSP2 ("Middle of the Road" or "Current Trends Continue"), while "Grand Deal" is analogous to SSP1 ("Sustainability" or "Taking the Green Road"). For the "SADC Integration" scenario, which generally portrays economic and social development between the BAU and Grand Deal scenarios, the study draws upon the International Futures model – a large-scale, long-term, integrated global modelling system using a general equilibrium framework and systems analysis (Hughes et al., 2009; International Futures, 2014). The features of the scenarios are described in Table 1, while narrative descriptions are provided in the Supplementary material.

3.2. Demand drivers and assumptions

The key drivers of electricity demand vary by socio-economic sector. In the residential sector, population growth, urbanisation and consumption per household are the main drivers across southern Africa. For industrial sectors, economic growth (e.g. gross domestic product), structure of the economy, and sectoral electricity intensity are most important. This section explains the choice of data sources for the demand drivers. The detailed assumptions are presented in Supplementary material for the article.

Population growth, urbanisation, and access to electricity determine the "activity level" for residential consumption (i.e. the social and economic activity that drives energy use). Because population growth (or more specifically, total fertility, as shown in the World Development Indicators (World Bank, 2014)) is inversely correlated with economic growth and human development, the population growth rates are lowest in the "Grand Deal" scenario. Population growth in the SADC Integration scenario is somewhat higher than the BAU scenario for some countries. This is because the International Futures (2014) Base Case (the source for this scenario), while projecting *total* population for the region that is between the values for the BAU and Grand Deal scenarios, includes dynamics for particular countries that are different to those of the other scenarios. Household size data (Central Statistics Office, 2010; Euromonitor International, 2013; IIASA, 2012) was used to estimate household numbers, which is the key driver of residential consumption.

The **access to electricity** in urban and rural areas for each country increases across all scenarios, but at different rates. The Grand Deal Scenario assumes that all countries reach the "sustainable energy for all" targets of 100% access in rural and urban areas by 2040, slightly later than the 2030 goal for this program (United Nations, 2015). The access levels in 2070 are between 70–100% for urban and 30–40% for rural households in the BAU and SADC Integrated scenarios respectively. In Botswana and South Africa, 100% of rural and urban households achieve access by 2070.

For residential demand, the electricity intensity parameter is the **average electricity consumption per household** with access to electricity. Where national residential demand was reported by the utility, this data is used for calculating average household consumption. For other countries, Mozambique is used as a proxy for average consumption based on broadly similar income levels. For the South

African demand analysis, electricity consumption is further disaggregated into end-uses using household data from Statistics South Africa (StatsSA, 2013) and electricity consumption from Eskom's annual report (Eskom, 2014). Relative electricity consumption of urban versus rural households is used to transform the national consumption data into rural and urban household average consumption value, based on more detailed research on South Africa residential demand (ERC, 2014, 2013).

For industrial sectors, the activity levels are determined by **sectoral and per capita GDP**, as well as **sectoral electricity intensity** (Price et al., 1998). Of the three economic modelling studies used for the GDP projections, only the International Futures model also considers the change in sectoral shares and this is used for the sectoral share changes over time. **Final electricity intensity** is either calculated from current sectoral electricity consumption and current sectoral GDP, or, where sectoral data is not available, estimated based on a country with similar income and calibrated to match total national demand (e.g. sum of all sectors).

Energy demand growth in the **transport sector** is driven by population growth and GDP for passenger and freight mobility, respectively. Electricity is only relevant for transportation in South Africa, where it is a significant share of consumption. The assumptions and detailed analysis of transportation energy use are taken directly from an analysis of the South African transport sector by Merven et al. (2012).

The **electricity load curve** is another important input for power system planning (Malik and Kuba, 2013). Based on previous experience modelling the South African power system (ERC, 2013), this study uses eight time slices over a day for all the four seasons. Hourly load data was available for South Africa, Mozambique, Swaziland and Zambia, so for these countries this hourly data was aggregated into the eight time slices. South Africa's load shape was used as a proxy for the other countries.

3.3. Supply assumptions

Power plant options are categorised as “existing plants”, “specific plants” (i.e. where there is a site specified as well as some technical and financial data) and “generic plants” (i.e. plants that could be located anywhere within a country and in any country). The detailed characteristics of each category of plants is presented in the [Supplementary material](#). More than 78% of the **existing power plant** capacity is in South Africa, and 75% of the region's existing capacity is thermal (of which almost 90% is coal-fired) (Table 2). In contrast to South Africa and Botswana, most of the other countries in the region rely primarily

on renewable power, all of which is hydropower outside of South Africa.

The pumped storage facilities in South Africa are a special case in the model because, although they are hydropower generators, the water is pumped into the storage reservoir at night using predominantly coal-fired power from the South African grid. This two-stage conversion requirement is not adequately represented in LEAP and therefore we created a proxy to simulate these generation components. These plants are therefore designated as coal-fired in the LEAP model, and their efficiency is the efficiency of large coal plants times the efficiency of the pumped storage unit (i.e. 0.38 for coal x 0.75 for storage), after Yang and Jackson (2011).

The total capacity of **specific future plants** is shown in Table 2 by country and type. Note that the South African Integrated Resource Plan (IRP) calls for more than 36 GW of new coal by 2030, but not all of this is a “specific” plant (DoE, 2013). Much of this growth is captured in the generic plant options in this study, because the IRP does not designate specific locations for all future plants. The largest individual specific project is the Grand Inga hydropower cascade in DRC, with an estimated total potential of more than 42 GW (Taliotis et al., 2014). Because this cascade must be developed in stages, in each scenario Grand Inga is commissioned in 3–7 GW sections of five years each, the difference being the starting date and total capacity. For the Grand Deal and SADC Integration scenarios, the full capacity is eventually realised, with the first phase in 2025 and 2030, respectively.

The **generic future plants** for generation included in the modelling cover a wide range of technologies from fossil fuels and nuclear to all major renewable technology types, although solar, wind, geothermal and biomass feature most prominently in regional plans. Not all plant types are available in each country (e.g. South Africa does not have large hydropower and some countries may have limited wind resources and biomass supplies).

A key feature of new emerging technologies is the potential for significant **capital cost reductions over time** from global experience (IEA, 2014b; IEA and OECD, 2000). For the Grand Deal scenario, the most optimistic cost reductions from the International Renewable Energy Agency (IRENA) study (Miketa and Merven, 2013) are used for the generic renewable energy technologies, with annual cost reductions of 0.5% to 4%. The BAU scenario uses the “pessimistic” scenario (i.e. no reductions), while the SADC Integration scenario uses the “reference case” assumptions.

The **technical characteristics of the fuels** used in the generation modelling are standard IEA assumptions (IEA, 2014a), except for coal, where the “other bituminous” coal has a net calorific value of 21 GJ/tonne (ERC, 2013), while the other fuels are from IEA data (IEA,

Table 2
Existing and specific proposal capacity by type and country, 2014 (MW).

Country	Existing capacity (MW)					Specific proposed capacity (MW)				
	Renewable	Thermal	Nuclear	Storage	Total	Renewable	Thermal	Nuclear	Storage	Total
Angola	776	1,075	0	0	1,851	5,165	500	0	0	5,665
Botswana	0	450	0	0	450	0	600	0	0	600
DRC	1,689	0	0	0	1,689	45,821	0	0	0	45,821
Lesotho	74	0	0	0	74	205	0	0	1,200	1,405
Malawi	334	46	0	0	380	845	100	0	0	945
Mozambique	2,175	399	0	0	2,574	6,545	2,225	0	0	8,770
Namibia	347	125	0	0	472	300	774	0	0	1,074
South Africa	1,640	44,582	1,860	1,580	49,662	10,663	9,656	0	1,332	21,651
Swaziland	171	0	0	0	171	137	300	0	0	437
Tanzania	562	684	0	0	1,246	2,262	2,340	0	0	4,602
Zambia	2,077	0	0	0	2,077	4,108	600	0	0	4,708
Zimbabwe	750	394	0	0	1,144	2,593	3,350	0	0	5,943
Total	10,595	47,755	1,860	1,580	61,789	78,644	20,445	0	2,532	101,621

Note: Renewable includes hydropower; storage=pumped storage plants; for detail of existing plants and sources, see Annex A. For detail of specific planned plants and sources, see Annex B.

2014a). GHG emission factors for fossil fuels are taken from the IPCC (IPCC, 2006). This study uses the same fossil **fuel costs** as the IRENA SAPP study (Miketa and Merven, 2013). Fuel prices increase over time in real terms. The change in oil and oil product prices is close to those values in the IEA's Energy Technology Perspectives (ETP) forecasts (IEA, 2014b). The coal price increases are higher, but this reflects the very low absolute prices of coal in southern Africa (i.e. almost half of the Organisation for Economic Cooperation and Development import price)⁴ and the move towards greater uniformity of prices over the coming decades. The IRENA SAPP study assumptions for natural gas prices include a steeper increase than the IEA forecast for North American and Japan. Biomass prices depend on the specific fuel type. In the countries with the largest sugar industries (i.e. Mozambique, Swaziland, Tanzania, and Zambia), the fuel is essentially free, but it is not available in the other countries.

According to the SAPP Coordination Centre, the short-term goal for the **planning reserve margin** is 10.2%, rising to 15% in 2020 (Maviya, 2014). This is used for all countries in the model to derive the capacity requirements to meet peak demand after 2015. Up to 2015, the actual regional reserve margin, as reported by SAPP, is used (Maviya, 2014; SAPP, 2012). Data on technical power **losses** in transmission and distribution and own use of electricity by utilities are both sourced from national utilities, electricity regulators, and the IEA (EdM, 2010; IEA, 2014c; LEC, 2015; NERSA, 2006; ZESCO, 2013a, 2013b), with Malawi and Swaziland being estimated from Mozambique.

Current electricity **trade flows** are as reported by SAPP member utilities in the SAPP annual reports, as well as in IEA statistics (IEA, 2014c, 2011; SAPP, 2014, 2013, 2010) and are presented in the [supplementary material](#). The SAPP data are reported per utility, however, and not per country, which requires adjustments for Mozambique and South Africa. For Mozambique, national exports are the net quantity of electricity sent to South Africa – which are the exports from Hidroelétrica de Cahora Bassa to South Africa less the re-import of this electricity via Motraco for the Mozal aluminium smelter (Mahumane and Mulder, 2015a, 2015b). Similarly, South Africa imports and exports are also net of the transmission of 8,466 GWh from northern Mozambique to Mozal in southern Mozambique via South Africa. Over the study period, the high-capacity transmission line from northern to southern Mozambique will be complete so it will not be necessary to use the South African grid to transport electricity to Mozal or other industrial developments in southern Mozambique.

Because transmission infrastructure is currently one of the main constraints on present trade in the SADC region, the scenarios with the highest foreign and regional investment rates would have the most trade because they imply a greater roll-out of higher capacity infrastructure. For the most aggressive growth scenario – Grand Deal – trade at the end of the study is based on the most optimistic forecast in the IRENA study (i.e. the “renewables promotion” scenario in Miketa and Merven, 2013). The SADC Integration Scenario is set to 60% of this trade, while BAU is 20% (which is still almost three times current trade flows). The exception to using the IRENA study inputs is the case of Angola and Lesotho, which both have large investments in capacity within “specific plants” that are for exports. Generation from this capacity (e.g. Kobong Pumped Storage Scheme for Lesotho and Kuanza Basin developments in Angola) has been used as the proxy for exports from these countries after their construction, so these do not vary across scenarios.

3.4. Calibration

Calibration with current data is important to validate the potential of any model to create accurate scenarios for the future. The details of the model calibration are presented in the [Supplementary material](#). Modelled national electricity demand was compared with reported consumption from SAPP annual reports and/or other utility and official sources. The modelled results have error discrepancies that are within less than 1% in all countries. Mozambique national demand deserves special attention, because the approach in this study is very different from the other similar studies. In the other studies, Mozambique's national demand is composed of that served by Electricidade de Moçambique (EDM) only and excludes that of the Mozal aluminium smelter, which is close to 20% of current national demand. The Mozal aluminium smelter demand is excluded in the SAPP reports for Mozambique because it is supplied from South Africa, but is included as national demand in this study because the demand is within the geographic boundaries of Mozambique.

For the supply model calibration, modelled capacity was compared with reported available capacity. LEAP does not directly distinguish between installed and available capacity, except through the “maximum availability” parameter,⁵ so the model should match the *available* power and not include installed plants that are not operational (these can be added back in future years). The installed and available capacities reported by SAPP (which can vary considerably due to some plants being out of service) are generally only the power plants owned by the SAPP member utilities. For this reason, the national utilities were asked to update national capacity, taking into consideration independent power producer and municipal generation, as well as any units that were out of service in 2010. Modelled capacity is very close to this updated utility capacity in all cases. For DRC, where updated capacity was not available, the modelled capacity is between the reported installed and available capacities. The ongoing renovation of Inga 2 and Inga 3 during this period meant that different units were not available in different years. Given these uncertainties, therefore, the calibration results show that the model matches the observed data well.

4. Results and discussion

The modelling results are presented in the following sections, starting with national and sectoral demand trends in [Section 4.1](#). This is followed by the results for difference types of electricity supply in [Section 4.2](#) – existing plants, specific new plants and generic plants that are added to optimise future supply to meet demand. System costs and GHG emissions are shown in [Sections 4.3 and 4.4](#), while discussion of the results and other studies in the region follow in [Section 4.5](#).

4.1. Demand

Depending on the scenario, total electricity demand for the SAPP region increases by 8–14 times over period from 2010 to 2070 ([Fig. 2](#)). In fact, by 2070, the rapidly growing countries of DRC, Mozambique, and Zambia reach demand levels that are, under most scenarios, higher than current South African demand. High growth rates in Tanzania put this country, in one scenario, at a level higher than current South African demand as well, while Malawi has rapid growth but it is still relatively small as a demand centre.

Although total demand for the region increases from the BAU to the SADC Integration scenarios and from there to the Grand Deal scenario, this is not necessarily true for each country. This is because the scenarios are not meant to be “low”, “middle” and “high”, but are,

⁴ Data from the EIA also shows South African coal prices being only a fraction of those in North America and Europe (http://www.eia.gov/countries/prices/coalprice_elec-gen.cfm).

⁵ For example, if a power plant with six units had two units out of service for the full year, the maximum availability for the entire plant could be 60% (or less, considering other planned and unplanned outages on the available units).

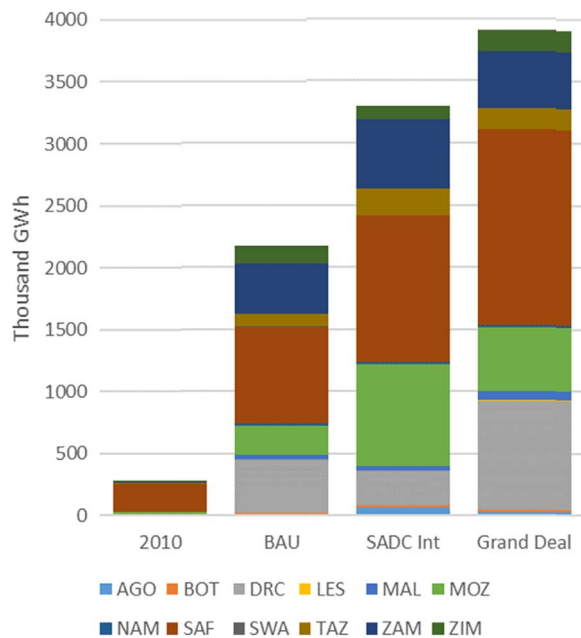


Fig. 2. Total electricity final demand by scenario, 2010 and 2070 (000 GWh).

instead, independent approaches to envisioning the future and do not simply change proportionally across each scenario. This is why Zambia, for example, has higher total demand in the SADC Integration scenario than under the Grand Deal scenario.

The sectors driving the growth in demand are primarily manufacturing and extractive (Fig. 3). In fact, demand from the extractive sector increases between 44 and 58 times from 2010 and 2070, both because of GDP growth and because of a rapid industrialisation of the region's economies. These combined effects lead to a 6.5–7.0% compound annual growth rate for extractive sector demand.

Changes in electricity demand are a product of the changes in activity level (e.g. population, GDP), energy intensity (e.g. industrial electricity intensity, consumption per household), and the structure of demand (e.g. share of GDP by sector, share of households with access, urbanisation). Activity level growth is clearly the main driver of demand growth, although changes in structure are also important for many countries (Fig. 4). The decreases in energy intensity, while

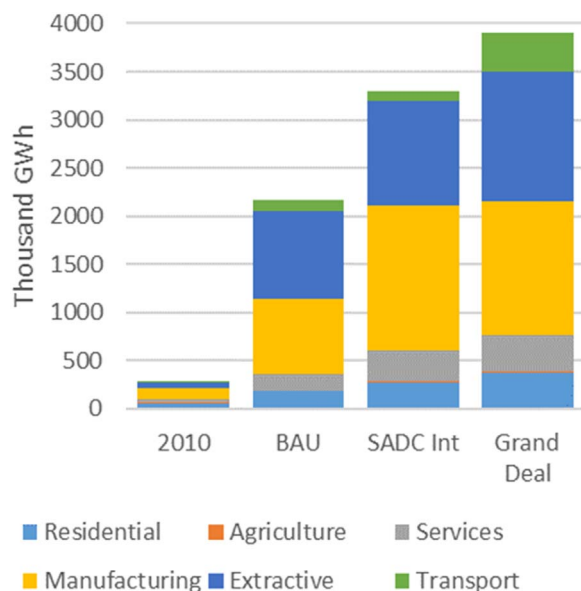


Fig. 3. Total electricity final demand by sector, 2010 and 2070.

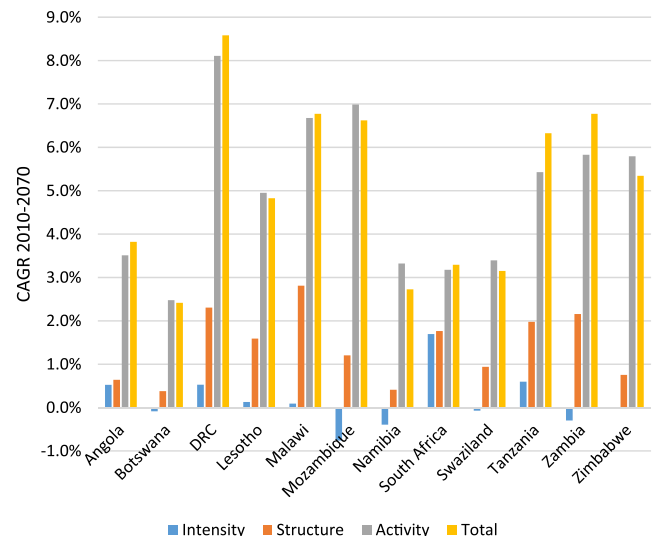


Fig. 4. Decomposition of changes in demand into activity level, structure, and intensity, under the Grand Deal scenario. Note: CAGR = Compound Annual Growth Rate; the columns labelled “Intensity”, “structure” and “activity” show the growth rates with the other two drivers of demand held constant.

important, are much smaller than the changes in activity levels.⁶

4.2. Supply

Existing plant capacity has increased from 2010 to 2015 due to expansions at existing facilities, largely coal plants in South Africa, bringing total regional capacity to just over 61.8 GW. These existing plants, however, are largely decommissioned during the study period, because only the large hydropower plants and the newest coal plants could last for more than 30 or 40 years (based on their estimated life from previous studies (ERC, 2014, 2013) and from inputs from SAPP Planning Sub-Committee members). This assumes that these large power plants that are used until the end of the study period will not only be maintained but also rehabilitated over time when necessary.

Growth in capacity from **new specific power plants** is fastest in Grand Deal scenario, and adds a total of nearly 80 GW to regional generation. The largest component of specific planned capacity is the Grand Inga Dam, which represents up to 42 GW in DRC. The key difference in the BAU scenario and the others is the lower total capacity from Grand Inga. In addition, the timeframe to 2070 is sufficiently long that some new plants in early years must be retired before the end of the study period. For example, the decommissioning of some of the solar and wind power in South Africa that is installed early in the study period leads to declines in total specific power plant capacity from 2040 to 2044. The South African expansion, which includes large investments in coal-fired power, is based in part on the building of Kusile and Medupi and in part on the generic plants used in optimisation.

Because the LEAP simulation and optimisation algorithms ensures that demand is always met by supply, the growth of **generic capacity** is far greater than the other two capacity types over the longer term, because of the very large increase in demand that must be met. As discussed above, by 2070 demand in the region could increase to between eight and fourteen times current levels. Capacity will therefore need to increase similarly (Fig. 5). Of the 500–1,400 GW capacity in 2070, only 60–80 GW will be from the remaining existing and specific plants.

The dramatic growth in demand and supply over the full study

⁶ Demand can increase even in the “intensity” scenario because decreases in industrial electricity intensity are offset by increases in residential consumption per household.

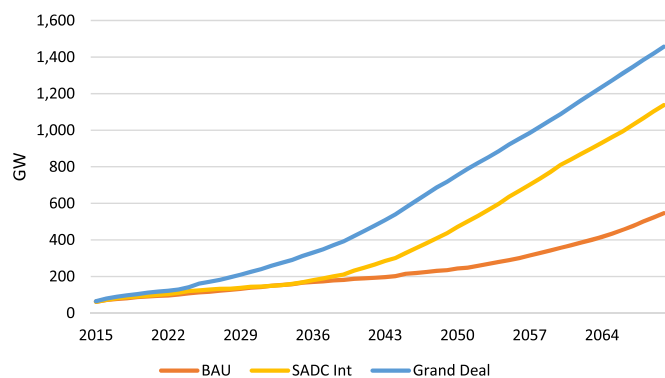


Fig. 5. Total capacity for all plants by scenario.

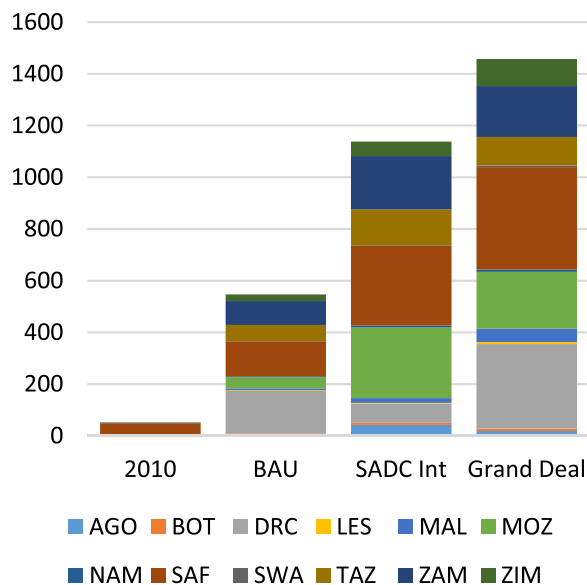


Fig. 6. Generation capacity by country and scenario, 2070 (GW).

period would result in a major shift in the regional power sector, both in terms of geography and fuel mix. By 2030 South Africa's share of regional capacity declines in all scenarios, as DRC, Mozambique and Zambia's power sectors rapidly expand (Fig. 6, Fig. 7). The fuel mix

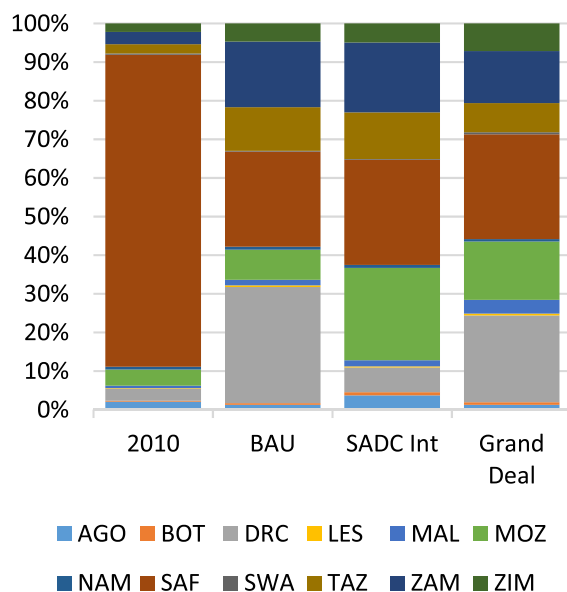


Fig. 7. Share of capacity by country and fuel, 2070.

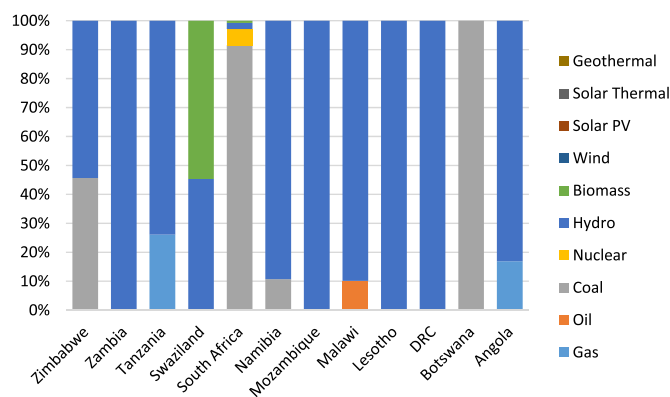


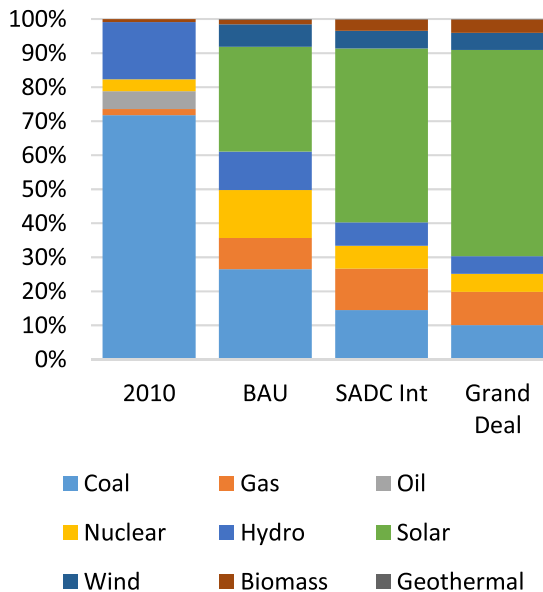
Fig. 8. Share of generation by fuel and country, 2010.

shifts away from coal and toward hydro in all scenarios, with a large share of solar in the Grand Deal scenario. By 2070 the shift is even more dramatic, with South Africa generating only a third of the region's power, and the fuel mix dominated by renewables – particularly utility-scale solar photovoltaics (Fig. 7). The share of renewables increases under the SADC Integration and Grand Deal scenarios because of the modelled declines in renewable energy capital costs under those scenarios, as discussed earlier.

To understand the implications of the scenarios for individual countries, a more disaggregated view of the scenario results is required. The starting fuel mix for each country in 2010 (Fig. 8) shows the domination of hydropower for most of the SAPP countries except Botswana, South Africa, and Zimbabwe – countries with rich coal resources that have been well developed.

Even under the BAU scenario, this fuel mix starts to shift by 2030, with Tanzania developing geothermal power, solar PV playing larger roles in several countries, coal starting in Zambia, Swaziland and Malawi, hydropower increasing in Namibia, and nuclear playing a larger role in the South African electricity mix. By 2070 under the BAU scenario, however, the role of coal has increased dramatically, because most large-scale hydropower resources are already exploited by 2040 or 2050 (Fig. 9). Nuclear provides the major source of South Africa's generation, while most other countries are dominated by coal. The exceptions are a large share of solar PV in DRC, Lesotho and Tanzania, and a significant share of wind in Botswana and Namibia.

Under the Grand Deal scenario, solar PV and wind have already



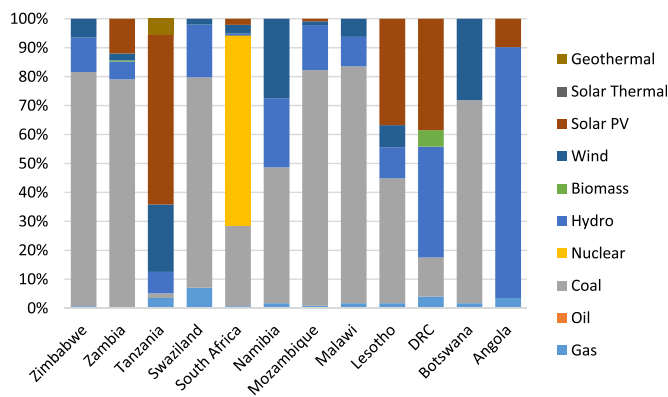


Fig. 9. Share of generation by fuel and country, 2070, BAU scenario.

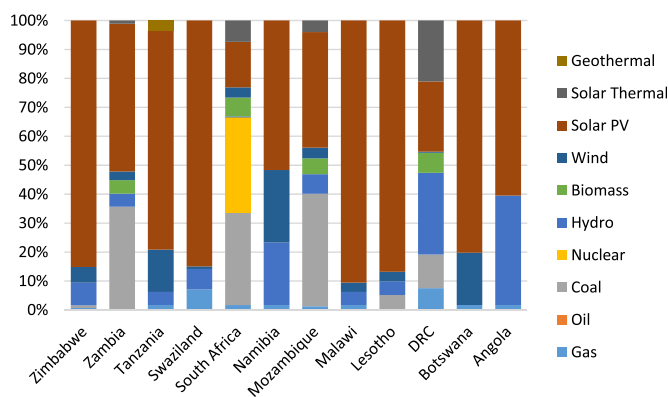


Fig. 10. Share of generation by fuel and country, 2070, Grand Deal scenario.

become much more prominent in the generation mix by 2030. Solar PV plays a significant role in Lesotho, Malawi, Mozambique, Namibia, South Africa, Swaziland and Zambia, while wind is a significant share of generation in Botswana, Malawi, Namibia and Zambia, displacing gas in some of those countries. The share of coal in South Africa is down to less than 60%. By 2070 under the Grand Deal scenario, declining capital costs makes solar PV a low-cost resource for most countries, and solar PV comprises the majority of generation in nine of the twelve countries (Fig. 10). Including biomass, solar thermal, wind and geothermal, renewable power is the majority of generation in all SAPP countries except South Africa, where renewable sources would be approximately 35% of generation.

4.3. System generation costs

The scope of this analysis is limited to the costs of generation, including all investment, maintenance and fuel costs required for generating electricity. Transmission and distribution costs are not included, nor are the cost of end-use appliances and energy-using equipment, but these are both small fraction of total costs. This is not only because of a lack of data for the other costs, but also because the focus of the research is how socio-economic development scenarios will impact generation options and costs. For capital costs, the amortised investment costs are included for each year, so this investment is spread out over the life of the plant, rather than all in one or two years. Although total costs increase in future years, the unit cost of generation for the entire region is relatively stable across all scenarios (Fig. 11). The cost structure shifts towards capital costs and, under the Grand Deal scenario, also toward variable operating and maintenance costs. Note that these costs do not include the “sunk costs” of the existing generation plants, even though the utilities may well have outstanding debt that must still be paid on those plants. This means that the total cost shown here underestimates utility cost in the early years of the

scenarios. In addition, these costs do not include revenue from exported electricity or the cost of imported electricity – they only include the costs of domestic generation. The shift away from feedstock fuel costs and towards capital and variable O & M costs in the Grand Deal scenario shows the transition to greater use of renewables.

4.4. Greenhouse gas emissions

Even with the shift towards renewable energy under most scenarios, total emissions for the region increase four to six times over the study period (Fig. 12). Only carbon dioxide emissions are included in the assessment, as these account for the vast majority of power station emissions. Because so much of the additional capacity in the Grand Deal scenario, when compared to the BAU and SADC Integration scenarios, is from renewable power, the GHG emissions of the three scenarios are largely the same. This means that the per unit emissions for the Grand Deal scenario is much lower than for the other two, given the much higher generation in this scenario and larger share from renewables and nuclear (Fig. 12).

4.5. Discussion

Because other studies of regional electricity supply and demand have generally only extended to 2025 or 2030, only a medium-term comparison of results is possible. Considering the two most recent shorter-term studies in the region and the SAPP forecasts (CEEZ, 2012; Miketa and Merven, 2013; SAPP, 2014), the modelling results in this study for demand are within the range of projected demand in 2025 (i.e. the last year of the SAPP forecasts) (Fig. 13). For the region as a whole, the modelled results are 450,000 to 610,000 GWh of final demand for electricity, while the three other sources range from 440,000 to 560,000 GWh. Given the optimistic assumptions used in the Grand Deal scenario (e.g. very rapid economic growth, full access to electricity by 2040), the fact that the high end of the modelled results exceed the other studies is reasonable. This comparison is important for understanding the longer-term results for this modelling exercise, because it means that the mid-term starting point (i.e. 2025) is also similar to what other regional initiatives considered. As discussed in Section 2, the methodology assumes that demand is not constrained by lack of available investment and so provides a more accurate picture of what investment is necessary to meet development needs.

There are some important differences in the demand results at a country level that also highlight the importance of bottom-up approaches to demand simulation. The RESAP (CEEZ, 2012) study, for example, used a constant growth rate for regional demand applied to each country (i.e. top-down approach), so this assumes that countries all grow at the same rate. Most of the SAPP forecasts, compiled from individual utilities, use almost the same growth rate in each year, even though this varies by country. The more detailed approach used in this study yields more conservative demand projections for Angola, Botswana and Zimbabwe, but more ambitious ones for Zambia. Note that the differences in Angola may also be related to the poor data availability, because the current data reported to SAPP imply very different energy intensities than other countries (suggesting that there could be underlying reporting problems). The major differences in Mozambique are because of the geographic definition of demand explained, so that Mozal and similar industrial demands, which are not part of Mozambique demand in other studies or SAPP reports, are part of the Mozambican national demand in this study. Because this study takes an approach that considers all demand within national boundaries as part of national demand, the Mozambique demand is much higher than forecast in SAPP reports, covering the full industrial base within the country. This is critical for policy makers in Mozambique, since it implies a different trajectory of power sector development to support industrial growth. For all of the countries, an important advance in this study is that the demand projections are

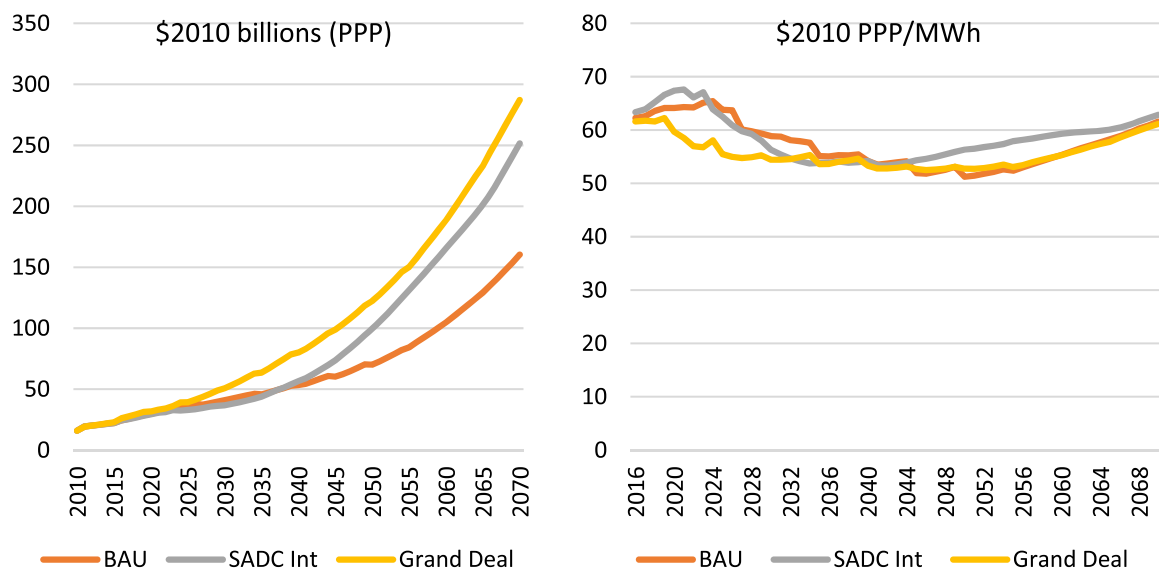


Fig. 11. Total and unit generation costs by scenario.

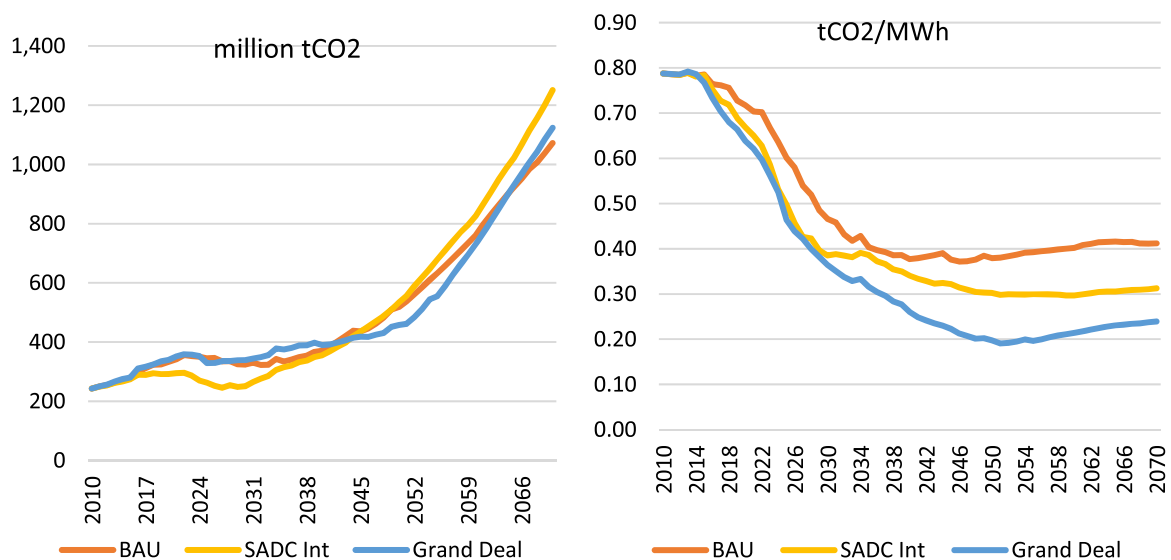


Fig. 12. Total and unit carbon dioxide emissions for each scenario.

based on scenarios covering national, regional and global development, which can be updated over time or used to test alternative assumptions about the future.

An important limitation in this analysis is not being able to include new transmission investments in the optimisation analysis. The modelling tools chosen do not allow this, and although the inputs drawn from the IRENA SAPP study were the result of a combined transmission and generation optimisation, the LEAP model is not able to dynamically adjust transmission flows in response to changing generation capacity and costs across countries. In addition, using time-of-day time slices (e.g. morning vs evening) in the generation modelling could provide additional insights on the value of trading because peak demand is not at the same time in each of the SAPP countries. Another important limitation is that the availability of large hydropower plants is largely based on historical river flow data, as provided by the regional utilities and previous regional research studies. There is mounting evidence, however, that water availability could change dramatically in the coming decades, with significant risks of lower water availability in the Zambezi River Basin (Kling et al., 2014; Spalding-Fecher et al., 2014; Stanzel and Kling, 2014). The next phase of this research will therefore examine the implications of

changes in availability of the major hydropower reservoirs along the Zambezi for regional and national electricity supply. In addition, while this study has focused on centralised power options for the region, there will also be a substantial increase in the need for technologies and capacity for integrating many smaller-scale distributed power systems into regional planning.

5. Conclusions and policy implications

The potential transformation of the supply sector presented in these scenarios would require a fundamental shift in resource use, grid management and infrastructure development in the region. For decades, South Africa, Botswana and, to a lesser extent, Zimbabwe, have built largely coal-fired plants, while the rest of the region has been dominated by hydropower, with more recent inroads from gas-fired power. As a result of the declining costs of renewable energy technologies – particularly solar photovoltaics and wind – the shift away from fossil fuels toward renewables could be dramatic over the coming decades. More importantly, this shift may not be driven by political or environmental reasons but by economic and financial ones. This follows a global trend highlighted by recent research showing renew-

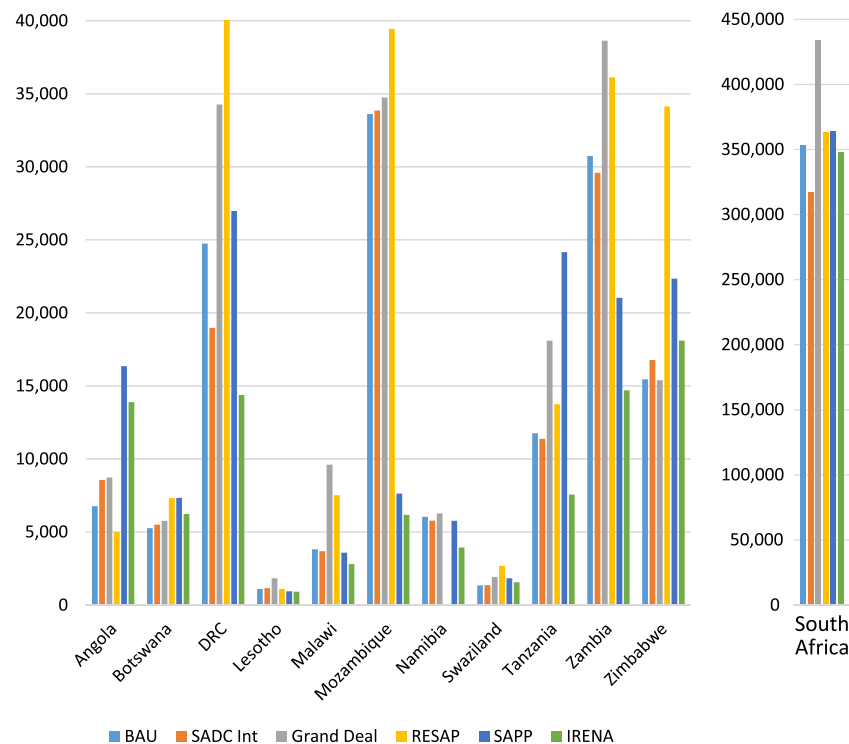


Fig. 13. Modelled demand versus other studies, 2025 (000 GWh).

able power not only achieving parity with traditional generation, but become less expensive and therefore comprising the majority of future capacity additions (see, e.g. BNEF, 2015; Randall, 2015). None of the short to medium term plans put forward for the region (e.g. the consolidated SAPP forecasts for capacity and demand) fully address the magnitude of this transition to a more diverse energy mix, and many ignore it entirely. The modelling approach in this study not only allows for a deeper and longer term analysis of supply, but also a more nuanced understanding of demand drivers (i.e. as opposed to the constant growth rate projections in some other studies).

This shift in the resource base for electricity generation will pose challenges for grid integration and balancing supply and demand across countries and load centres. In fact, the lack of availability of adequate transmission capacity, the ability to balance dispatchable and non-dispatchable sources of supply, and the lack of availability of cost-effective storage could be the most important limitations on realising an optimal expansion plan. The only major storage in the region other than the two major hydropower reservoirs at Kariba and Cahora Bassa currently is the pumped storage facilities in South Africa (1.58 GW), but even with another 1.2 GW in South Africa and 1.3 GW in Lesotho planned, these are far smaller than the additional solar and wind power investments over the next two decades. Major advances in chemical or other storage technologies are therefore important to realising the magnitude of solar power expansion. In the short-to-medium term, exploring the possibility of “banking” energy from non-dispatchable resources (e.g. solar PV, wind) in large hydropower reservoirs in the region (i.e. beyond the dedicated pumped storage facilities) could be one important strategy for this new supply regime. This will require a greater focus on commissioning new transmission capacity, but also an analysis of whether modifying the operating rules at major regional hydropower reservoirs could comprise their other functions (e.g. flood control, ecological flow releases).

Historically the development of transmission capacity, and the resulting trade in electricity, has been constrained by the political and economic realities of the region. The tension between the potential for savings from trade versus the additional costs of maintaining national self-sufficiency has been a recurring theme in many previous studies

(Economic Consulting Associates, 2009; Graeber and Spalding-Fecher, 2000; Nexant, 2007; Rowlands, 1998), and requires both political and technical solutions to address security of supply concerns. Regional infrastructure plans have identified political commitment, as well as strong institutions, as one of the keys to greater cooperation (SADC, 2012). To date, trade in the short term market has been thin and the SAPP Coordination Centre has not directly facilitated longer term contracts and negotiations. For the largest source of demand in the region – South Africa – imported electricity has played a very limited role in future planning (DoE, 2013; Resnick et al., 2012). There are signs that the politics could be shifting, however, for several key reasons. The first is the political commitment and financing available for major regional energy infrastructure projects. At the continental level, the Program on Infrastructure Development in Africa (PIDA) initiative of the African Union and the New Partnership for African Development (NEPAD) represents a new level of political commitment to mobilize several hundred billion dollars for infrastructure projects, including regional energy infrastructure projects to facilitate greater energy trade (Mandelli et al., 2014). In Southern Africa specifically, the World Bank launched the “SAPP Program for Accelerating Transformational Energy Projects” to provide transaction advisory and other services to address the technical and economic barriers large and complex transmission projects that are necessary for greater regional cooperation. Finally, the growing scarcity of water in the largest economies of Southern Africa, which will be exacerbated by climate change, could force a political shift, as the relative political power of countries with more water and other unexploited natural resources grows (Turton, 2016).

In terms of the specific scenarios analysed in this report, although they do represent an increase in trade, they do not require major changes in national energy security policies for most countries. Over the longer term, net imports (i.e. imports minus exports) as a share of national demand are relatively small in all countries in this simulation (Fig. 14). Countries with negative net imports are net exporters, which are almost all countries except Zambia and South Africa, as well as Malawi in 2070. By 2070, imports for these countries are less than 5% of national demand in all scenarios. However, in the medium term, net

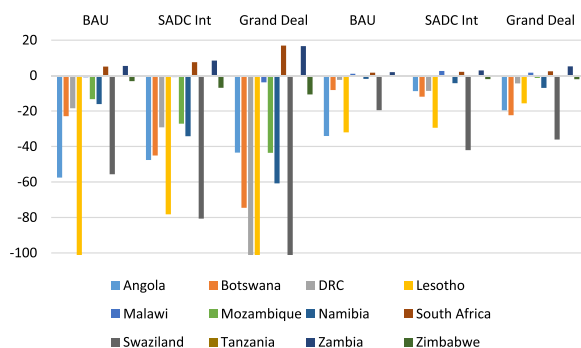


Fig. 14. Net imports as a share of national final demand by country (%). Note: values less than -100% - DRC Grand Deal = -102, Lesotho BAU = -125, Lesotho Grand Deal = -139, Swaziland Grand Deal = 197.

imports for South Africa and Zambia by 2030 could be a significant share of domestic demand under the SADC Integration and Grand Deal scenarios, and so the political and financing drivers discussed above would be important to realizing these scenarios.

The differences in medium-term demand projections for some countries between this study and the SAPP compilation of utility forecasts also suggests that the lack of national planning frameworks driven by bottom-up demand analysis could lead to significant over- or under-investment. Capacity among utilities for developing detailed bottom-up demand analysis and supply optimisation analysis varies widely. Several SAPP utilities reported to the authors that they have no planning model for their power sector, and this reflects the weak institutional capacity in utilities and regulators across Africa (Eberhard et al., 2011; Eberhard and Shkaratan, 2012). A dedicated capacity-building program for the regional utilities could address this, but only if it combines both management level buy-in on greater cooperation with greater technical skill, data sharing and collaborative planning among member utilities. The program should also utilize a “learning by doing” approach, where utility staff learn the modelling tools through application to real SAPP data during a series of workshops, with technical backstopping between these meetings. The experience in this study, as well as interacting with stakeholders during the process of research, also highlighted some of the benefits of using LEAP as a tool for national and regional planning in Southern Africa. First, LEAP has the ability to combine simulation analysis with optimisation analysis, so that the current SAPP Planning Framework can be used as an input to the analysis. This has important implications for the policy impact of this approach because decision makers contributing elements of the SAPP plan have a greater stake in the outputs. Second, in workshops presenting the work at different stages, stakeholders noted the user-friendliness and accessibility of the model, particularly users from outside the major power companies who may not have significant energy modelling expertise. In fact, user-friendliness is one of the key reasons why LEAP has been adopted by thousands of organisations in more than 190 countries (SEI, 2015). The SADC RESAP study mentioned above also used LEAP, and has been reviewed by not only SADC staff and management but also the energy ministers. Because some of the barriers to greater regional cooperation relate to institutional and technical capacity, the accessibility of the analytical tool is relevant to the policy impact of the analysis. Third, the affordability of the tools is relevant to policy impact, because this influences whether regional institutions can continue to update and use the analysis in the future. LEAP is available without cost to public sector and academic users in developing countries, and is supported by a professional team at no cost to these users. These last two aspects greatly increase the potential for sustainability in the application and maintenance of the tool.

The institutional capacity to manage grid integrated, accelerated transmission investments and cooperative management of resources

must also be further strengthened. As one example, the construction and operation of more large-scale hydropower plants within the Zambezi River Basin and export of power will increase the importance of coordinated operation across countries. Currently a Joint Operating Technical Committee (JOTC), with representatives from Zambia, Zimbabwe and Mozambique, exists to provide limited coordination of the Kariba Dam and Cahora Bassa Dam. The entry into force of the “Agreement on the Establishment of the Zambezi Watercourse Commission” (the ZAMCOM Agreement) in 2011 and, more importantly, the establishment of a permanent Secretariat in 2014, provides the foundation upon which to develop greater coordination and efficiency of resource utilization across the SAPP countries that are part of the Zambezi River Basin. However, there is still no formal institutional cooperation between the SAPP Coordination Centre and ZAMCOM, or other regional water management institutions.

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Appendix A. Supplementary information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.enpol.2016.10.033](https://doi.org/10.1016/j.enpol.2016.10.033).

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